Chap 2 II Integration Theory

1. Lebesgue Integral

Def II.1.1 The Canonical form: of a simple function $\varphi := \sum_{k=1}^{M} a_k \chi_{E_k}$

$a_k \neq a_j$ $(\forall k \neq j)$ are all values φ could take E_k are disjoint

Def II.1.2. Let φ be a simple function. the Lebesgue integral of $\varphi := \sum_{k=1}^{M} c_k m(E_k)$

Denoted by $\int_{\mathbb{R}^d} \varphi(x)\, dx$

If E is a measurable subset of $\mathbb{R}^d$. then $\int_E \varphi(x)\, dx := \int_{\mathbb{R}^d} \varphi(x)\, \chi_E(x)\, dx$

Prop II.1.3. φ simple

(1) $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$ be any representation.

Then $\int \varphi = \sum_{k=1}^{N} a_k\, m(E_k)$

(2) $\int (a\varphi + b\psi) = a\int \varphi + b\int \psi$

(3) E, F disjoint. $m(E), m(F)$ finite, then $\int_{E\cup F} \varphi = \int_E \varphi + \int_F \psi$.

(4) $\varphi \leq \psi$ then $\int \varphi \leq \int \psi$

(5) φ is simple, then $|\varphi|$ is simple, and $\left|\int \varphi\right| \leq \int |\varphi|$

Proof: Only (1),

Let $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$. First assume that E_k are disjoint.

For each distinct non-zero value a among $\{a_k\}$, define $E_a' = \bigcup E_k$.

where the union is taken over k. such that $a_k = a$. then $m(E_a') = \sum m(E_k)$

Obviously, $E_a' \cap E_b' = \phi$. $(a \neq b)$. $\Rightarrow \int \varphi = \sum a\, m(E_a') = \sum_{k=1}^{N} a_k\, m(E_k)$

Now let $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$ Consider $E_j^* = E_1' \cap E_2' \cap \dots \cap E_N'$.

E_k' is either E_k or E_k^c. then $E_k = \bigcup E_j^*$ $(E_k \supseteq E_j^*)$. $\bigcup_{j=1}^{2^N-1} E_j^* = \bigcup_{k=1}^{N} E_k$

let $a_j^* = \sum a_k$. for all k s.t. $E_k \supseteq E_j^*$ then E_j^* are disjoint,

a_j^* are distinct. $\Rightarrow \int \varphi = \sum a_j^* m(E_j^*) = \sum_k \sum_{E_k \supseteq E_j^*} a_k\, m(E_j^*) = \sum a_k\, m(E_k)$ □.

Def II.1.4. f is measurable, the support set of f: $\operatorname{supp}(f) := \{x \mid f(x) \neq 0\}$.

f is supported on E if $f(x) = 0$ whenever $x \notin E$

Lemma II 1.5 f is a bounded function on a set E of finite measure. If $\{\varphi_n\}_{n\geq 1}$ is any sequence of simple functions bounded (supported) by M, supported on E and with $\varphi_n(x) \to f(x)$ a.e. x, then (i) $\lim_{n\to\infty} \int \varphi_n$ exists

(ii) If $f = 0$ a.e., then $\lim_{n\to\infty} \int \varphi_n = 0$

proof: (i) Given $\varepsilon > 0$ — By Egorov's thm $\exists$ closed $A_\varepsilon \subseteq E$ s.t. $m(E - A_\varepsilon) < \varepsilon$, $\varphi_n \to f$ uniformly on A_ε.

Let $I_n = \int \varphi_n$. then $|I_n - I_m| \leq \int_E |\varphi_n - \varphi_m| dx$

$$= \int_{A_\varepsilon} |\varphi_n - \varphi_m| dx + \int_{E-A_\varepsilon} |\varphi_n - \varphi_m| dx$$

$$\leq 2 m(E)\varepsilon + 2M\varepsilon \quad \text{when } m \text{ and } n \text{ is large enough}$$

Thus $\{I_n\}$ converges.

(ii). If $f = 0$ a.e. then $|I_n| \leq m(E)\varepsilon + 2M\varepsilon \Rightarrow \lim_{n\to\infty} I_n = 0$ $\square$.

Def II.1.6. f is a bounded function supported on E, $m(E) < \infty$

Then define $\int f(x)\,dx = \lim_{n\to\infty} \int \varphi_n(x)\,dx$.

$$\int_E f(x)\,dx = \int f(x)\,\chi_E(x)\,dx$$

Thm II.1.7. (Bounded Convergence Theorem). $\{f_n\}$ bounded by M, supported on E. $m(E) < \infty$. and $f_n(x) \to f(x)$ a.e. x $(n \to \infty)$ Then f is measurable, bounded, supported on E for a.e. x, (*)
and $\int f_n \to \int f$ $(n \to \infty)$

Proof: (*) is obvious. By Egorov thm: find A_ε

$$\int |f_n - f|\,dx \leq \int_{A_\varepsilon} |f_n(x) - f(x)|\,dx + \int_{E-A_\varepsilon} |f_n(x) - f(x)|\,dx$$

$$\leq m(E)\varepsilon + 2M\, m(E - A_\varepsilon) \quad \text{for all large enough } n$$

$$\Rightarrow \int |f_n - f|\,dx = 0 \quad (n \to \infty) \Rightarrow \int f_n \to \int f \quad (n \to \infty) \quad \square$$

Rmk (1) that is $\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$

(2) $f \geq 0$, bounded, supported on E, $m(E) < \infty$ and $\int f = 0$. then $f = 0$ a.e. x

Let $E_k = \{x \in E \mid f(x) > \frac{1}{k}\}$. then $\frac{1}{k}\chi_{E_k}(x) \leq f(x) \Rightarrow \frac{1}{k} m(E_k) \leq 0 \Rightarrow m(E_k) = 0$

Since $\{x \mid f(x) > 0\} = \bigcup_{n\geq 1} E_n$, we have $f = 0$ a.e. x $\square$

Def II.1.8. f is non-negative, measurable define $\int f(x)dx = \sup_g \int g(x)dx$, where $0 \le g \le f$, g bounded, supported on a set of finite measure.

If $\sup < \infty$ then we say f is Lebesgue integrable

Prop. II.1.9. $f, g \ge 0$, integrable,

(i) $a, b > 0 \Rightarrow \int(af+bg) = a\int f + b\int g$

(ii) $E \cap F = \phi$, $\Rightarrow \int_{E\cup F} f = \int_E f + \int_F f$.

(iii) $0 \le f \le g \Rightarrow \int f \le \int g$

(iv) g is integrable, $0 \le f \le g \Rightarrow f$ is integrable

(v) f is integrable, then $f < \infty$ a.e. x.

(vi) $\int f = 0$, then $f(x) = 0$ a.e. x.

proof: (i) $a = 1 = b$, if $\varphi \le f$, $\psi \le g$. φ, ψ bounded, supp on E, $m(E) < \infty$

then $\varphi + \psi \le f+g$. then $\varphi+\psi$ bounded, supp on E', $m(E') < \infty$

$\Rightarrow \int f + \int g \le \int (f+g)$

Let η be bounded, supp on E, $m(E) < \infty$, $\eta \le f+g$.

$\eta_1 = \min\{f, g\}$, $\eta_2 = \eta - \eta_1$. then $\eta_1 \le f$, $\eta_2 \le g$.

$\Rightarrow \int \eta = \int(\eta_1+\eta_2) = \int \eta_1 + \int \eta_2 \le \int f + \int g$. Take superum of η

$\Rightarrow \int(f+g) \le \int f + \int g \Rightarrow \int f + \int g = \int(f+g)$ □

(Fatou)

Lemma II.1.10. $\{f_n\} \ge 0$, measurable, If $\lim_{n\to\infty} f_n(x) = f(x)$ a.e. x. then $\int f \le \liminf_{n\to\infty} \int f_n$

proof: $\forall g$, $0 \le g \le f$. bounded, supp on E, $m(E) < \infty$, let $g_n = \min\{g, f_n\} \le f_n$

$g_n \to g$ a.e. $x \Rightarrow \int g_n \to \int g$ (By bounded convergence thm)

$\int g_n \le \int f_n$ Take the liminf, we have $\int g \le \liminf_{n\to\infty} \int f_n$.

Take the superum of g, we have $\int f \le \liminf_{n\to\infty} \int f_n$

Def II.1.11. $f_n \nearrow f$ refers to $f_n \le f_{n+1}$ a.e. x, all $n \ge 1$. and $\lim_{n\to\infty} f_n(x) = f(x)$ a.e. x.

Similarly, $f_n \searrow f$.

Coro II.1.12. (Monotone convergence theorem) $f_n \ge 0$, $f_n \nearrow f$. Then $\lim_{n\to\infty} \int f_n = \int f$.

proof: $f_n \le f$. $\Rightarrow \int f_n \le \int f$ $\forall n \ge 1$. $\Rightarrow \limsup_{n\to\infty} \int f_n \le \int f$.

By Fatou's thm, we immediately have $\lim_{n\to\infty} \int f_n = \int f$. □.

Coro II.1.13 $\sum_{k=1}^{\infty} a_k(x)$ where $a_n(x) \geq 0$ measurable. Then $\int \sum_{k=1}^{\infty} a_k(x)\,dx = \sum_{k=1}^{\infty} \int a_k(x)\,dx$

If $\sum_{k=1}^{\infty} \int a_k(x)\,dx$ is finite, then $\sum_{k=1}^{\infty} a_k(x)$ converges a.e. x.

The general case:

Def II.1.14. f is Lebesgue integrable if $|f|$ is integrable

Let $f^+ = \max\{f, 0\}$. $f^- = \max\{-f, 0\}$ $\Rightarrow$ $f = f^+ - f^-$

Define $\int f = \int f^+ - \int f^-$.

Prop. II.1.15. The integral of Lebesgue integrable functions is linear, additive, monotonic, and satisfies the triangle inequality.

Proof (skipped)

Prop. II.1.16. f is integrable on $\mathbb{R}^d$. Then $\forall \varepsilon > 0$.

(i) $\exists B$, $m(B) < \infty$. s.t. $\int_{B^c} |f| < \varepsilon$.

(ii) $\exists \delta > 0$. s.t. $\int_E |f| < \varepsilon$. whenever $m(E) < \delta$.

Proof, W.L.O.G $f \geq 0$. (i) Let $f_N(x) = f(x)\,\chi_{B_N}(x)$. B_N is the ball of radius N at the origin

Then $f_N \nearrow f$. By monotone convergence theorem: $\lim_{N\to\infty} \int f_N = \int f$.

$\Rightarrow \exists N > 0$. s.t. $0 < \int f - \int f\,\chi_{B_N} < \varepsilon$ $\Rightarrow$ $0 < \int f\,\chi_{B_N^c} < \varepsilon$ $\Rightarrow$ $\int_{B_N^c} f < \varepsilon$

(ii). $\forall \varepsilon > 0$ $\exists N > 0$. s.t. $0 < \int f - \int f_N < \frac{\varepsilon}{2}$. Choose $\delta > 0$. s.t. $N\delta < \frac{\varepsilon}{2}$. $\forall\, m(E) < \delta$.

$$\int_E f = \int_E (f - f_N) + \int_E f_N$$

$$\leq \int (f - f_N) + N m(E) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \square$$

Thm II.1.17. (Lebesgue dominated convergence theorem) f_n measurable, $f_n(x) \to f(x)$ a.e. x

If $|f_n(x)| \leq g(x)$, $g(x)$ is integrable. then $\int f_n \to \int f$ $(n \to \infty)$

Proof: $\forall N \geq 0$. 令 $E_N := \{x \mid |x| \leq N,\ g(x) \leq N\}$, $\exists$ large N. s.t. $\int_{E_N^c} g < \varepsilon$.

$\Rightarrow \int_{E_N} f_n$ is bounded. supp on a E, $m(E) < \infty$

By bounded convergence thm, $\int_{E_N} |f_n - f| < \varepsilon$. $\forall n$ large

$\Rightarrow \int |f_n - f| \leq \int_{E_N} |f_n - f| + 2\int_{E_N^c} g \leq 3\varepsilon$.